



ABBOTSLEIGH

AUGUST 2003
YEAR 12
ASSESSMENT 4
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators may be used
- A table of standard integrals is provided with this paper
- All necessary working should be shown in every question

Total marks - 84

- Attempt Questions 1-7
- All questions are of equal value

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 Attempt Questions 1-7
 All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- | | | |
|-----|--|---|
| (a) | Solve $\frac{4}{x-1} \geq 1$ | 3 |
| (b) | A is the point $(-2, -1)$ and B is the point $(1, 5)$. Find the coordinates of the point Q which divides AB externally in the ratio $5:2$. | 2 |
| (c) | Given $f(x) = \tan^{-1}(\sin x)$ find $f'(\pi)$ | 2 |
| (d) | Prove $\frac{1 + \sin x - \cos x}{1 + \sin x + \cos x} = \tan \frac{x}{2}$ | 2 |
| (e) | Find the exact value of $\int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{(3-4x^2)}}$ | 3 |

End of Question 1

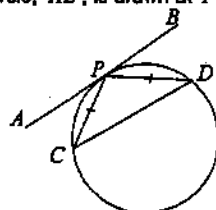
Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve the equation $2\sin^2 \theta = \sin 2\theta$ for $0 \leq \theta \leq 2\pi$

2

- (b) PC and PD are equal chords of a circle.
A tangent to the circle, AB , is drawn at P .



Copy the diagram into your answer booklet and prove that AB is parallel to CD . 2

(c) (i) Find $\int \frac{x}{x+9} dx$

2

(ii) Evaluate $\int_0^4 x\sqrt{x^2+9} dx$ using the substitution $u = x^2 + 9$

3

(d) (i) Sketch $y = |x+1|$

1

(ii) Using your graph, or otherwise, solve $|x+1| > -2x$ for x

2

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) For the polynomial $P(x) = x^3 - kx^2 - x + 2$

(i) Find the value of k if $x-1$ is a factor of $P(x)$.

1

(ii) Hence factorise $P(x)$ completely.

2

(b) Find the term which is independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^9$

2

(c) For the function $f(x) = 4\sin^{-1}(x-2)$

(i) Evaluate $f\left(1\frac{1}{2}\right)$

1

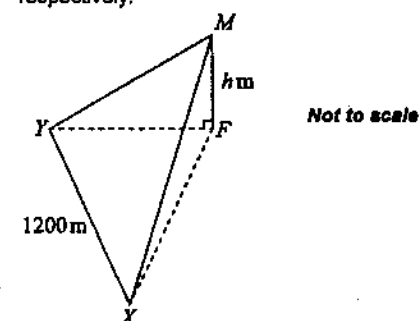
(ii) Sketch $y = f(x)$ clearly indicating the domain and range.

2

(iii) Find $\int_1^3 4\sin^{-1}(x-2) dx$

1

- (d) In the diagram, Point X is due south and point Y is due west of the foot, F , of a mountain. From X and Y , the angles of elevation of the top of the mountain M are 35° and 43° respectively.



If X and Y are 1200 metres apart, show that the height, h metres, of the mountain is given by $h = 1200(\tan^2 55^\circ + \tan^2 47^\circ)^{-\frac{1}{2}}$ and evaluate h .

3

End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

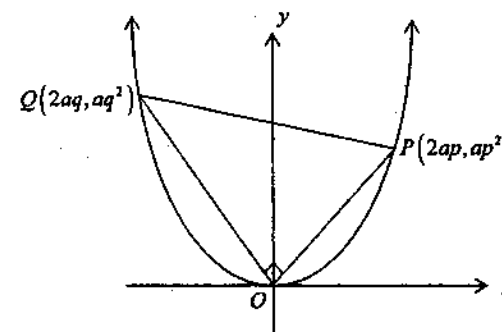
- (a) (i) Sketch the graph of $y = \cos x$, $-\pi \leq x \leq \pi$ and use this graph to show that $\cos x + x = 0$ has only one solution. 2
- (ii) Use Newton's method with a first approximation of $x = -1$ to find a second approximation to the root of $\cos x + x = 0$. 2
- (b) The inside of a vessel used for water has the shape of a solid of revolution obtained by the rotation of the parabola $9y = 8x^2$ about the y -axis. The depth of the vessel is 8 cm.
- (i) Prove that the volume of water h cm from its base is $\frac{9}{16}\pi h^2$. 1
- (ii) If water is poured into the vessel at a rate of $20 \text{ cm}^3/\text{sec}$, find the rate at which the level of water is rising when the vessel is half full. 3
- (c) Use the Principle of Mathematical Induction to prove that $2^{3n} - 3^n$ is divisible by 5 for all positive integers n . 4

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) In the diagram, PQ is a variable chord of the parabola $x^2 = 4ay$. It subtends a right angle at the vertex O . Let p and q be the parameters corresponding to the points P and Q respectively.



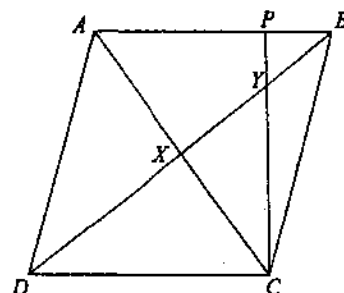
- (i) Show that the equation of the tangent to $x^2 = 4ay$ at P is $y - px + ap^2 = 0$. 1
- (ii) Hence, write down the equation of the tangent at Q , and then find R , the point of intersection of the two tangents drawn at P and Q . 2
- (iii) Find the gradients of OP and OQ and hence prove $pq = -4$. 2
- (iv) Show that the locus of R , the point of intersection of the two tangents drawn at P and Q is $y = -4a$. 1

- (b) By considering $f(x) = (1+x)^n$ in $\int_0^1 f(x) dx$, prove that

$$\sum_{r=0}^n \frac{1}{r+1} \binom{n}{r} = \frac{2^{n+1} - 1}{n+1}$$

Question 5 continues on page 7

- (c) $ABCD$ is a rhombus whose diagonals intersect at X . The perpendicular CP from C to AB cuts BD at Y .



Not to scale

Copy the diagram into your writing booklet and prove that B, P, X, C are concyclic.

3

End of Question 5

- (a) Find $\int \sin^2 x \cos^2 x \, dx$ 2
- (b) A particle moves in a straight line so that its velocity after t seconds is $v \text{ ms}^{-1}$ and its displacement is x .
- (i) Given that $\frac{d^2x}{dt^2} = 2x^3 - 10x$ and that initially $v = 0$ when $x = -1$, find v in terms of x . 3
- (ii) Explain why this motion can only exist between $x = -1$ and $x = 1$. 2
- (iii) Describe briefly what would have happened if the initial conditions were $v = 0$ when $x = 0$. 1
- (c) In a colony of 400 ants the number, N , diseased at time, t , is given by $N = \frac{400}{1 + ke^{-400t}}$ where k is a constant and t is time in years. (Assume one year is 365 days.)
- (i) If at time $t = 0$ one ant was infected, after how many days will half the colony be infected? 3
- (ii) Show that eventually all the ants will be infected. 1

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) A particle is projected from a point on level ground with a speed of $V \text{ ms}^{-1}$ and angle of projection, α . Assume that acceleration due to gravity is $g \text{ ms}^{-2}$ and that there is no air resistance.

- (i) Show that the horizontal and vertical displacements, x and y , of the particle in metres from the point of projection at time t seconds after projection are given by

$$x = Vt \cos \alpha \quad \text{and} \quad y = Vt \sin \alpha - \frac{1}{2}gt^2 \quad 2$$

- (ii) Show that the greatest height of the particle is $\frac{V^2 \sin^2 \alpha}{2g}$ 1

- (iii) Show that the range of the particle is $\frac{V^2 \sin 2\alpha}{g}$ 1

- (iv) Two particles are projected from the same point on level ground with the same speed $V \text{ ms}^{-1}$ and with angles of projection α and $90^\circ - \alpha$ respectively.

The greatest heights the two particles reach are h_1 and h_2 respectively.

Show that, for any α , $h_1 + h_2 = \frac{1}{2}R$ where R is the maximum range. 3

- (b) A_n and B_n are two series given by

$$A_n = 1^2 + 5^2 + 9^2 + 13^2 + \dots + (4n-3)^2$$

$$B_n = 3^2 + 7^2 + 11^2 + 15^2 + \dots \quad \text{for } n = 1, 2, 3, \dots$$

- (i) Find the n th term of B_n . 1

- (ii) If $S_{2n} = A_n - B_n$, prove that $S_{2n} = -8n^2$. 2

- (iii) Hence, or otherwise, evaluate

$$101^2 - 103^2 + 105^2 - 107^2 + \dots + 2001^2 - 2003^2 \quad 2$$

End of Paper

Question 1

(a) $\frac{4}{x-1} \geq 1$ $x \neq 1$

$$4(x-1) \geq (x-1)^2$$

$$(x-1)^2 - 4(x-1) \leq 0$$

$$(x-1)[x-1-4] \leq 0$$

$$(x-1)(x-5) \leq 0$$

$$1 < x \leq 5$$

(b) A(-2, -1) B(1, 5) S:-2

$$x = \frac{5 \times 1 - 2 \times -2}{5 - 2} \quad y = \frac{5 \times 5 - 2 \times -1}{5 - 2}$$

$$= 3 \quad = 9$$

$$Q(3, 9)$$

(c) $f(x) = \tan^{-1}(\sin x)$

$$f'(x) = \frac{\cos x}{1 + \sin^2 x}$$

$$f'(\pi) = \frac{\cos \pi}{1 + \sin^2 \pi} = -1$$

(d) LHS = $\frac{1 + \sin x - \cos x}{1 + \sin x + \cos x}$

$$= \frac{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \frac{1+t^2+2t-1+t^2}{1+t^2+2t+1-t^2}$$

$$= \frac{2t^2+2t}{2+2t}$$

$$= \frac{2t(t+1)}{2(t+1)} = t = \tan \frac{x}{2}$$

(e) $\int_0^{\sqrt{3}/2} \frac{dx}{\sqrt{3-4x^2}} = \int_0^{\sqrt{3}/2} \frac{dx}{2\sqrt{\frac{3}{4}-x^2}}$

$$= \frac{1}{2} \left[\sin^{-1} \frac{2x}{\sqrt{3}} \right]_0^{\sqrt{3}/2}$$

$$= \frac{1}{2} (\sin^{-1} 1 - \sin^{-1} 0)$$

$$= \frac{\pi}{4}$$

Question 2

(a) $2 \sin^2 \theta = \sin 2\theta$, $0 \leq \theta \leq 2\pi$

$$2 \sin^2 \theta = 2 \sin \theta \cos \theta$$

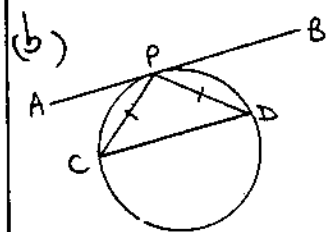
$$\sin^2 \theta - \sin \theta \cos \theta = 0$$

$$\sin \theta (\sin \theta - \cos \theta) = 0$$

$$\sin \theta = 0, \sin \theta = \cos \theta$$

$$\tan \theta = 1$$

$$\theta = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$$



$\angle PCD = \angle PDC$ given equal chords
 $\triangle PDC$ isosceles
 with equal base angle

$\angle BPC = \angle PCD$ angle between tangent
 + chord = $\angle$ in alternate
 segment

$$\therefore \angle BPC = \angle PDC$$

The alternate angles are equal

$\therefore AB$ parallel to CD

(c) (i) $\int \frac{x}{x+9} dx = \int \frac{x+9-9}{x+9} dx$

$$= \int \left(\frac{x+9}{x+9} - \frac{9}{x+9} \right) dx$$

$$= \int \left(1 - \frac{9}{x+9} \right) dx$$

$$= x - 9 \ln(x+9) + C$$

(ii) $\int_0^4 x \sqrt{x^2+9} dx$ $u = x^2+9$

$$= \frac{1}{2} \int_0^4 \sqrt{x^2+9} \cdot 2x dx$$

$$= \frac{1}{2} \int_9^{25} \sqrt{u} du$$

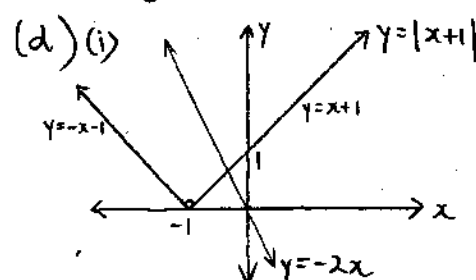
$$= \frac{1}{2} \int_9^{25} u^{\frac{1}{2}} du$$

$$= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_9^{25}$$

$$= \frac{1}{3} \left((\sqrt{25})^3 - (\sqrt{9})^3 \right)$$

$$= \frac{1}{3} (125 - 27)$$

$$= \frac{98}{3}$$



(ii) One pt. of intersection

Solve $x+1 = -2x$

$$3x = -1$$

$$x = -\frac{1}{3}$$

$$\therefore |x+1| > -2x \text{ when } x > -\frac{1}{3}$$

Question 3

(a) $P(x) = x^3 - kx^2 - x + 2$

(i) $x-1$ is a factor $\therefore P(1) = 0$

$$0 = 1 - k - 1 + 2$$

$$k = 2$$

(ii) $P(x) = x^3 - 2x^2 - x + 2$

$$= x^2(x-2) - (x-2)$$

$$= (x-2)(x^2-1)$$

$$= (x-2)(x-1)(x+1)$$

(b) $T_{k+1} = {}^9C_k \left(\frac{2}{x}\right)^k (x^2)^{9-k}$

$$= {}^9C_k 2^k x^{-k} x^{18-2k}$$

$$= {}^9C_k 2^k x^{18-3k}$$

term indept. of x ie x^0

$$\therefore 0 = 18 - 3k$$

$$k = 6$$

7th term is independent of x

$$T_7 = {}^9C_6 2^6$$

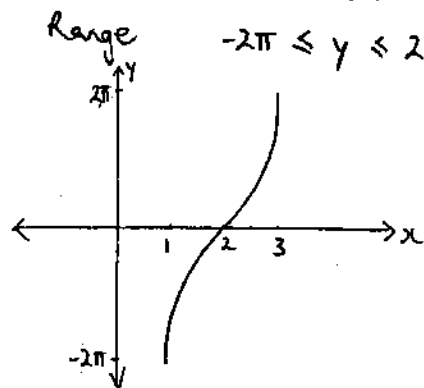
$$= 2^8 \times 3 \times 7$$

$$= 5376$$

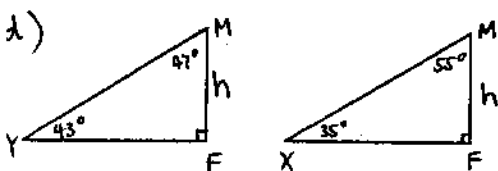
c) $f(x) = 4 \sin^{-1}(x-2)$

(i) $f(1\frac{1}{2}) = 4 \sin^{-1}(-\frac{1}{2})$
 $= 4 \times -\frac{\pi}{6}$
 $= -\frac{2\pi}{3}$

(ii) Domain $-1 \leq x-2 \leq 1$
 $1 \leq x \leq 3$
 Range $-2\pi \leq y \leq 2\pi$



(iii) $\int_1^3 4 \sin^{-1}(x-2) dx = 0$



In $\triangle MYF$ $\tan 47^\circ = \frac{YF}{h}$
 $YF = h \tan 47^\circ$

In $\triangle MXF$ $\tan 55^\circ = \frac{XF}{h}$
 $XF = h \tan 55^\circ$

In $\triangle YFX$
 $1200^2 = YF^2 + XF^2$
 (Pythagoras)

$1200^2 = h^2 \tan^2 47^\circ + h^2 \tan^2 55^\circ$
 $h^2 (\tan^2 47^\circ + \tan^2 55^\circ) = 1200^2$

$h^2 = \frac{1200^2}{(\tan^2 47^\circ + \tan^2 55^\circ)}$

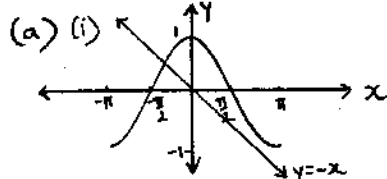
$h = \frac{1200}{\sqrt{\tan^2 47^\circ + \tan^2 55^\circ}}$

$h = 1200 (\tan^2 47^\circ + \tan^2 55^\circ)^{-\frac{1}{2}}$

$h = 671.915 \dots m$

$h \approx 671.9 m$ (to 1 dp)

Question 4



$\cos x + x = 0 \Rightarrow \cos x = -x$

Consider $y = \cos x$ & $y = -x$

Graphs intersect at one pt. only

$\therefore \cos x + x = 0$ has only one soln.

(ii) $f(x) = \cos x + x$

$f'(x) = -\sin x + 1$

$x_1 = -1$ (in radians)

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $= -1 - \frac{\cos(-1) - 1}{1 - \sin(-1)}$
 $= -0.75036 \dots$

2nd approximation is

$x = -0.75$ (to 2 dp)

b) (i) $V = \pi \int_0^h \frac{9y}{8} dy$
 $= \frac{9\pi}{8} \left[\frac{1}{2} y^2 \right]_0^h$
 $= \frac{9\pi}{8} \times \frac{1}{2} h^2 - 0$
 $= \frac{9\pi h^2}{16}$

(ii) $\frac{dV}{dt} = 20$ $\frac{dV}{dh} = \frac{18\pi h}{16} = \frac{9\pi h}{8}$

Find h when $\frac{1}{2}$ full.

Full $\Rightarrow h = 8 \text{ cm}$ $V = 36\pi$

$\frac{1}{2}$ full $\Rightarrow V = 18\pi = \frac{9\pi h^2}{16}$

$h^2 = 32$

$h = 4\sqrt{2}$

$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$

$20 = \frac{9\pi h}{8} \times \frac{dh}{dt}$ $h = 4\sqrt{2}$

$\frac{20 \times 8}{9\pi \times 4\sqrt{2}} = \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{40}{9\pi\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$= \frac{20\sqrt{2}}{9\pi} \text{ cm/sec}$

(c) $2^{3n} - 3^n$ divisible by 5

Show true for $n=1$

$2^3 - 3^1 = 8 - 3$

$= 5$ which is $\div$ by 5

Assume true for $n=k$

ie assume $2^{3k} - 3^k = 5M$ where M is a positive integer

Show true for $n=k+1$

$2^{3(k+1)} - 3^{k+1} = 2^{3k+3} - 3^{k+1}$

$= 2^3 \times 2^{3k} - 3 \times 3^k$

$= 2^3 (5M + 3^k) - 3 \times 3^k$

using assumption

$= 40M + 8 \times 3^k - 3 \times 3^k$

$= 40M + 5 \times 3^k$

$= 5(8M + 3^k)$ which is divisible by 5

$\therefore$ true for $n=k+1$ if true for $n=k$

Since true for $n=1$ it is also true for $n=1+1=2$ & thus true for $n=2+1=3$ and so on for all positive integers n .

Question 5

(a) (i) $y = \frac{x^2}{4a} \Rightarrow y' = \frac{x}{2a}$

at $P(2ap, ap^2)$ $m_T = \frac{2ap}{2a} = p$

eg. tangent $y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y - px + ap^2 = 0$

(ii) at Q tangent eq. $y - qx + aq^2 = 0$

$$y = px - ap^2, \quad y = qx - aq^2$$

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q)$$

$$x = a(p + q) \quad p \neq q$$

$$y = ap(p + q) - ap^2$$

$$y = apq$$

$$R(a(p + q), apq)$$

$$(iii) m_{OP} = \frac{ap^2 - 0}{2ap - 0} = \frac{p}{2}$$

$$\text{Similarly } m_{OQ} = \frac{q}{2}$$

$$OP \perp OQ \therefore m_{OP} \times m_{OQ} = -1$$

$$\text{i.e. } \frac{pq}{4} = -1 \Rightarrow pq = -4$$

(iv) at R $x = a(p + q)$

$$y = apq$$

$$\text{from (iii)} \quad pq = -4 \Rightarrow y = -4a$$

$\therefore$ locus of R is $y = -4a$

$$(b) f(x) = (1+x)^n$$

$$\begin{aligned} \int_0^1 (1+x)^n dx &= \left[\frac{(1+x)^{n+1}}{n+1} \right]_0^1 \\ &= \frac{2^{n+1}}{n+1} - \frac{1}{n+1} \\ &= \frac{2^{n+1} - 1}{n+1} \end{aligned}$$

Also

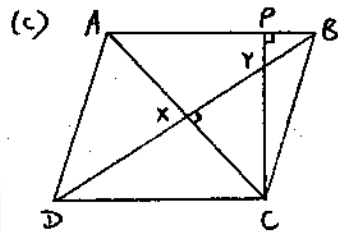
$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{r}x^r + \dots$$

$$\int_0^1 (1+x)^n dx = \left[\binom{n}{0}x + \frac{1}{2}\binom{n}{1}x^2 + \frac{1}{3}\binom{n}{2}x^3 + \dots + \frac{1}{r+1}\binom{n}{r}x^{r+1} + \dots \right]_0^1$$

$$= \binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \dots + \frac{1}{r+1}\binom{n}{r} + \dots + \frac{1}{n+1}\binom{n}{n} - 0$$

$$= \sum_{r=0}^n \frac{1}{r+1} \binom{n}{r}$$

$$\therefore \sum_{r=0}^n \frac{1}{r+1} \binom{n}{r} = \frac{2^{n+1} - 1}{n+1}$$



$\angle BPC = 90^\circ$ given

$\angle BXC = 90^\circ$ given rhombus; diagonals bisect each other at right angles.

Interval BC subtends equal angles on the same side of it at points P and X

$\therefore B, C, X, P$ are concyclic

Question 6

$$(a) \int \sin^2 x \cos^2 x dx$$

$$= \int (\sin x \cos x)^2 dx$$

$$= \int \left(\frac{1}{2} \sin 2x \right)^2 dx$$

$$\begin{aligned} \int \sin^2 x \cos^2 x dx &= \frac{1}{4} \int \sin^2 2x dx \\ &= \frac{1}{4} \int \frac{1}{2} (1 - \cos 4x) dx \\ &= \frac{1}{8} \left(x - \frac{1}{4} \sin 4x \right) + C \\ &= \frac{x}{8} - \frac{1}{32} \sin 4x + C \end{aligned}$$

$$(b) (i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3 - 10x$$

$$\frac{1}{2} v^2 = \frac{x^4}{2} - 5x^2 + C$$

$$v = 0 \text{ when } x = -1$$

$$0 = \frac{1}{2} - 5 + C$$

$$C = \frac{9}{2}$$

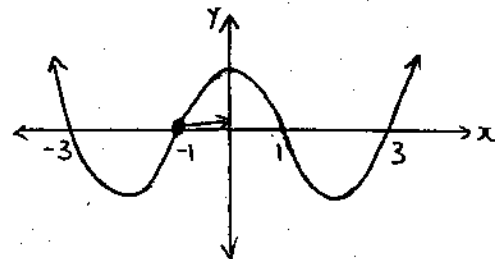
$$\frac{1}{2} v^2 = \frac{x^4}{2} - 5x^2 + \frac{9}{2}$$

$$v^2 = x^4 - 10x^2 + 9$$

$$v = \pm \sqrt{x^4 - 10x^2 + 9}$$

$$(ii) v^2 = (x^2 - 9)(x^2 - 1)$$

$$v^2 = (x - 3)(x + 3)(x - 1)(x + 1)$$



Starts at $x = -1$ and $v^2 \geq 0$

$\therefore$ only movement from $x = -1$ is to $x = 1$ + back. Exists between

(iii) If $x = 0$ then $a = 0$

With $v = 0$ and $a = 0$ there is no acceleration + no velocity $\therefore$ particle would not move.

(c) (i) $t = 0 \quad N = 1$

$$1 = \frac{400}{1 + ke^0}$$

$$1 + k = 400$$

$$k = 399$$

find t when $N = 200$

$$200 = \frac{400}{1 + 399e^{-400t}}$$

$$1 + 399e^{-400t} = 2$$

$$399e^{-400t} = 1$$

$$e^{-400t} = \frac{1}{399}$$

$$-400t = \ln \frac{1}{399}$$

$$t = -\frac{1}{400} \ln \frac{1}{399}$$

$$= 0.0149 \dots \text{ years}$$

$$\doteq 5.5 \text{ days}$$

(ii) as $t \rightarrow \infty$

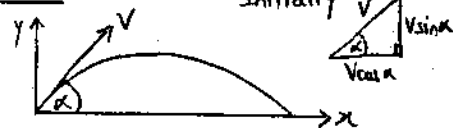
$$e^{-400t} \rightarrow 0$$

$$\therefore N \rightarrow \frac{400}{1 + 0} = 400$$

i.e. all the ants will be infected.

Question 7

(i)



Initially V
 $V \sin \alpha$
 $V \cos \alpha$

Horizontal

Vertical

$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

$$\dot{x} = C_1$$

$$\dot{y} = -gt + C_2$$

$$\text{when } t=0 \quad \dot{x} = V \cos \alpha$$

$$\text{when } t=0 \quad \dot{y} = V \sin \alpha$$

$$\therefore C_1 = V \cos \alpha$$

$$\therefore C_2 = V \sin \alpha$$

$$\dot{x} = V \cos \alpha$$

$$\dot{y} = V \sin \alpha - gt$$

$$x = Vt \cos \alpha + C_3$$

$$y = Vt \sin \alpha - \frac{1}{2}gt^2 + C_4$$

$$\text{when } t=0 \quad x=0$$

$$\text{when } t=0 \quad y=0$$

$$\therefore C_3 = 0$$

$$\therefore C_4 = 0$$

$$x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - \frac{1}{2}gt^2$$

(ii) greatest ht when $\dot{y} = 0$

$$V \sin \alpha = gt$$

$$t = \frac{V \sin \alpha}{g}$$

$$y = V \times \frac{V \sin \alpha}{g} \sin \alpha - \frac{1}{2}g \left(\frac{V \sin \alpha}{g} \right)^2$$

$$= \frac{V^2 \sin^2 \alpha}{2g}$$

(iii) range $\Rightarrow y=0$

$$0 = t \left(V \sin \alpha - \frac{1}{2}gt \right)$$

$$t=0, \quad t = \frac{2V \sin \alpha}{g}$$

$$x = V \times \frac{2V \sin \alpha}{g} \cos \alpha$$

$$= \frac{V^2 \times 2 \sin \alpha \cos \alpha}{g} = \frac{V^2 \sin 2\alpha}{g}$$

(iv) max. range when $\sin 2\alpha = 1$

$$\therefore R = \frac{V^2}{g}$$

$$h_1 = \frac{V^2 \sin^2 \alpha}{2g} \quad h_2 = \frac{V^2 \sin^2 (90-\alpha)}{2g}$$

$$h_1 + h_2 = \frac{V^2}{2g} (\sin^2 \alpha + \cos^2 \alpha)$$

$$= \frac{V^2}{2g}$$

$$= \frac{1}{2} R$$

(b) (i) $B_n = 3^2 + 7^2 + \dots + (4n-1)^2$
n-th term $(4n-1)^2$

$$(ii) S_{2n} = A_n - B_n$$

$$= 1^2 + 5^2 + \dots + (4n-3)^2$$

$$- 3^2 - 7^2 - \dots - (4n-1)^2$$

$$S_{2n} = (1^2 - 3^2) + (5^2 - 7^2) + (9^2 - 11^2)$$

$$+ \dots + [(4n-3)^2 - (4n-1)^2]$$

$$= (1-3)(1+3) + (5-7)(5+7) + \dots + (9-11)(9+11)$$

$$+ \dots + (4n-3-4n+1)(4n-3+4n-1)$$

$$= -2(4) - 2(12) - 2(20) - \dots - 2(8n-4)$$

$$= -2(4 + 12 + 20 + \dots + (8n-4))$$

$$= -2 \left(\frac{n}{2} (8 + (n-1)8) \right)$$

$$= -n(8 + 8n - 8)$$

$$= -8n^2$$

(iii) $101^2 - 103^2 + 105^2 - \dots + 2001^2 - 2003^2$
starts from $n=26$ to $n=501$

$$S_{501} - S_{25} = -8(501)^2 + 8(25)^2$$

$$= -2003008$$